

3° Longitudinal Periodicity in the UNESCO World Heritage Corpus: Enrichment in Domed Monuments and Harmonic Cluster Asymmetry

Seth Tenenbaum 

¹Independent Scholar, USA |

Correspondence: Seth Tenenbaum (seth@fourthtemple.com)

Received: **Revised:** **Accepted:**

Funding: This research received no external funding.

Keywords: longitude periodicity | spatial statistics | Levant | UNESCO World Heritage | 3-degree harmonics | domed architecture | archaeoastronomy | monument distribution | Mount Gerizim

ABSTRACT

The UNESCO Cultural and Mixed World Heritage corpus ($N = 1011$) shows a weak but statistically detectable periodic component in longitude consistent with a 3° model phased on Mount Gerizim (35.269°E). Three threshold-free tests reach nominal significance under Holm correction ($k = 3$): dome von Mises mixture, Spearman cluster-asymmetry, and phase-rotation null. The signal locates a ~35°E corridor, not a unique anchor. The Wikidata stupa and OWTRAD corpora show the same enrichment under the same frame but share geographic structure with UNESCO. The pattern appears only under this representation, not under invariant spatial analyses, and does not reflect intrinsic periodicity. Results are exploratory; pipeline fully reproducible.

1 | Introduction

This paper examines how a coordinate-dependent scoring framework can produce reproducible phase-localized alignment patterns in a geographically clustered corpus. The corpus is UNESCO Cultural and Mixed World Heritage sites ($N = 1011$). A global parameter sweep over frequency and phase selects a high-scoring frame near 3° spacing and the ~35°E corridor (§4.1, §5.2). All subsequent analyses are conditional on this selected frame. Because the frame is chosen by optimizing over the data, results at this frame reflect post-selection and should not be read as independent statistical evidence. Gerizim provides one historically documented point within the identified corridor (§2.2). Within this frame, the data show weak but reproducible departures from spatial uniformity. The same pattern is absent at non-optimal anchors and is not recovered as a spectral peak under invariant periodogram analysis.

Three diagnostic checks point in the same direction: dome von Mises mixture, Spearman cluster asymmetry, and phase-rotation null (Table 3; $p_{\text{adj}} = 0.004, 0.002, 0.047$). Effects are small ($A+$ rate 13.1% vs. 10% null; $\rho = -0.1076$); the anomaly localizes

the ~35°E corridor, not a unique anchor. This is compatible with 3° spacing but does not establish it (§2.2, §4.1).

The observed spatial regularity is concentrated in the longitude range where Eurasian cultural site density is highest (the Eurasian high-density corridor, often referred to as the Silk Road region) and is consistent with the geographic clustering of culturally related sites. The Wikidata Q180987 stupa inventory and OWTRAD route network show similar enrichment patterns under the same reference frame, but share geographic and cultural structure with UNESCO. Their agreement is cross-corpus consistency, not independent confirmation (§7.3).

2 | Background

2.1 | Prior Work on Spatial Periodicity and Sacred Landscapes

Statistical evaluation of spatial periodicity in monument distributions has a long and contested history (Thom 1967, Kendall 1974,

Freeman 1976). Probability-based frameworks for archaeoastronomical alignments have been advanced by Magli (Magli 2013 2016) and Ruggles and Silva (Ruggles 1999 2015, Silva 2020); the ICOMOS-IAU thematic study (Ruggles & Cotte 2010) provides institutional context for World Heritage analysis.

2.2 | Reference Phase and Candidate Anchors

The global anchor sweep (§5.2) places the $\sim 35^\circ\text{E}$ corridor among the highest-scoring reference phases under the harmonic-deviation score, with marginal discovery-corrected support: both Gerizim and Jerusalem place the $x.25^\circ\text{E}$ band in the global top 2.3% of 36,000 trial anchors, while Mecca ($p \approx 0.2825$, ns), Mt. Meru ($p \approx 0.3956$, ns), and Mt. Kailash ($p \approx 0.3956$, ns) yield no enrichment. Harmonic deviations are computed from Mount Gerizim (35.269°E), which lies within that corridor. The Samaritan *tabbur ha-aretz* (“navel of the earth”) tradition (Purvis 1968, MacDonald 1964, Crown 1989) provides independent historical motivation for this choice. Tests using Gerizim reflect properties of the selected corridor, not of any uniquely identified meridian. Gerizim is not inscribed (UNESCO tentative ref. 5706, DMS E35 16 08); Jerusalem is inscribed and self-excluded when used as anchor.

3 | Data and Methods

3.1 | Corpus Construction

The UNESCO World Heritage XML dataset (UNESCO World Heritage Centre 2024) contains 1248 inscribed nominations. UNESCO designates each site *Cultural* (criteria i to vi), *Natural* (criteria vii to x), or *Mixed*; sites with cultural heritage alongside natural values receive Mixed and are included here. Retaining Cultural and Mixed (-235 Natural-only) and excluding 2 entries lacking geocoordinates (both linear route networks) yields $N = 1011$ sites.

3.2 | Geodetic Framework

The reference anchor is Mount Gerizim (35.269°E ; not inscribed, §2.2). Jerusalem is inscribed and scored as a regular corpus site. For sensitivity sweeps, Gerizim is added synthetically so each anchor self-excludes symmetrically.

For a site at longitude λ , the deviation from the nearest 3° harmonic is $\delta = |\lambda - 35.269| \bmod 3^\circ$ (folded to $[0, 1.5^\circ]$). The tier classification is a fixed angular criterion. Kilometer labels (e.g. $\lesssim 16.6$ km for Tier-A+) are equatorial approximations ($\delta \times 111$ km/ $^\circ$). No latitude correction is applied. Kilometer-scale distances should be interpreted as approximate: UNESCO coordinates are site centroids, and precision below ~ 1 km is not meaningful.

The geometric null for Tier-A+ is $2 \times 0.15^\circ / 1.5^\circ = 10\%$ (assuming uniform longitudes). A Gaussian KDE gives a similar expected A+ count under longitude clumping (101.1 ± 9.6 expected A+). The primary diagnostics use simulation-based nulls (§5.4).

3.3 | Statistical Framework

The choice of $(\omega, \phi) = (3^\circ, \text{Gerizim})$ follows from a global parameter sweep. Define $T = \max_{\omega, \phi} S(\omega, \phi)$ where S is the harmonic-deviation enrichment score; T is calibrated by permuting site longitudes and recomputing, correcting for the full search. Global scans over ϕ (36,000 anchors) and unit scans over ω both maximize at $3^\circ / 35^\circ\text{E}$ (§4.1, §5.2; anchor-sweep permutation $p = 0.0753^\dagger$). S tests lattice proximity under a specified frame, not intrinsic periodicity; the two are epistemically distinct (§6.3). Three threshold-free diagnostics below characterize alignment at this fixed frame (see §3.5):

Test 1 Two-component von Mises/uniform mixture fit to the dome sub-corpus ($N = 90$), mean fixed at Gerizim phase, assessed by parametric bootstrap.

Test 2 Spearman rank correlation between harmonic node mean deviation and site count, full corpus ($N = 1011$), permutation null.

Test 3 Phase-rotation null: the reference grid is shifted by a uniform random offset in $[0^\circ, 3^\circ)$; tests whether the Gerizim phase minimizes the full-corpus mean harmonic deviation.

All three point in the same direction (diagnostic p -values in Table 3; Holm-adjusted at $k = 3$). All primary diagnostics operate on the same harmonic-deviation representation and should be interpreted as complementary summaries of a single constructed feature, not as independent evidence. These p -values quantify structure within the constructed frame and are not evidence against spatial randomness in an invariant sense. Tier labels (A++, A+, A) are descriptive vocabulary only and do not enter any diagnostic statistic (§3.4).

3.4 | Tier Classification

Sites are assigned to tiers by their angular deviation δ from the nearest 3° harmonic node. The three nested A-windows span the same range of proximities examined in the multi-scale enrichment analysis (§4.5): 4% null (≈ 6.66 km) through 20% null (≈ 33 km), covering the full interval over which the four corpora show enrichment. These windows are not identified as peaks; they are fixed proximity bands used as descriptive vocabulary for summarising harmonic proximity. No primary test statistic depends on a tier cutoff.

Tier A++ $\delta \leq 0.06^\circ$ ($\lesssim 6.66$ km); $P_0 = 4\%$.

Tier A+ $\delta \leq 0.15^\circ$ ($\lesssim 16.6$ km); $P_0 = 10\%$.

Tier A $\delta \leq 0.3^\circ$ ($\lesssim 33$ km); $P_0 = 20\%$.

Tier B Middle zone, between harmonics and inter-harmonic midpoints.

Tier C, C–, C–– Symmetric mirror of A, A+, and A++, measured from the nearest inter-harmonic midpoint.

3.5 | Test Populations

Primary threshold-free diagnostics (Tests 1 to 3)

Tests 1 to 3 are described in full in §3.3.

The only additional detail specific to this section is the keyword population for Test 1: 7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site’s name, XML description, and extended OUV text ($N = 90$). The context-validated

subset ($N = 83$) restricts to sites where the keyword appears in an architectural context and is reported as a robustness check (Test 5x).

Supporting enrichment tests (Tests 4 to 6)

Tests 4 to 6 count sites falling within a fixed proximity window of the nearest harmonic node and test whether the rate exceeds a geometric null. Three nested windows are reported (Tiers A++, A+, A; null rates 4%, 10%, 20%; widths 0.06° , 0.15° , 0.3° on $[0^\circ, 1.5^\circ)$). No primary diagnostic depends on a tier cutoff (see §3.4).

Test 4 (full-corpus enrichment). Full corpus ($N = 1011$), no keyword filter. Binomial test of A+ rate vs. $p_0 = 10\%$ null.

Test 5 (dome enrichment). Dome keyword subset ($N = 90$). Fisher exact and binomial tests of A+ and A++ rates.

Test 6 (mound evolution, sensitivity). Extends the dome subset with *tumulus*, *tumuli*, *barrow*, *barrows*, *kofun* ($N = 110$); not in the primary diagnostic family.

Descriptive and post-hoc tests

Test 7 (temporal gradient). Cochran-Armitage trend test on inscription year. Descriptive only.

Test 8 (founding classifier, post-hoc). Keyword classifier; no weight in the primary evidential chain (Appendix C.2).

3.6 | Multiple-Comparisons Protocol

Primary diagnostic family ($k = 3$, Holm step-down). Tests 1 to 3 are all threshold-free; they form the primary diagnostic family. The adjustment calibrates these summaries within the selected representation. It is not a confirmatory familywise test of discovery.

Supporting enrichment tests (Tests 4 and 5). Tests 4 and 5 use a pre-specified proximity window and are not in the primary diagnostic family. Results are in the FDR audit (Table 3; §4.8).

Other. Test 6 is a sensitivity variant of Test 5. Test 7 (temporal gradient) is descriptive. Test 8 is post-hoc. Religion subgroup analyses (§4.4) and diagnostic Tests A to D (§4.3) are not Bonferroni-corrected.

3.7 | Simulation Null Models

Three simulation approaches (100,000 trials each), a within-dome bootstrap stability check, and a restricted geographic-concentration null are described with their results in §5.4.

3.8 | Computational Reproducibility

All statistics are generated by an automated pipeline; every script writes to a shared results store and LaTeX macros are regenerated without manual editing. Scripts, audit files, and reproduction instructions are catalogued in Supplementary S1; see also §C.2.

4 | Results

Effects are weak in absolute terms (full-corpus A+ rate 13.1% against a 10% null; Spearman $\rho = -0.1076$).

4.1 | Spacing Specificity: Is 3° Discriminated from Nearby Intervals?

The 3° spacing is the highest-scoring parameter from the unit sweep (§3.3); metrological context is noted in §2.2. Within the canonical range examined, no nearby random spacing reproduces the observed tier-bin peak: a longitude-preserving permutation of the dome sub-corpus (10000 trials) yields $p = 0.0999$ at A^+ ($\sim$) and $p = 0.1728$ at A^{++} (ns). Restricting to morphologically unambiguous spherical forms (stupa, tholos, $n = 22$) gives $p = 0.0239$ (*) at the A tier.

A $\pm 0.3\%$ shift from exactly 3° collapses this peak in at least one population, so the result is best read as compatibility with a 3° model rather than precise recovery of a 3° unit (§5.1).

The 1.5° and 6° spacings produce similar tier-bin counts but are mathematical consequences of the 3° structure (shared harmonic nodes). Neither constitutes an independent signal. Candidate intervals at 4° and 8° (§6.2) are similarly arithmetic consequences of the 3° grid. Results support relative optimality of 3° within the tested family. It is a representative optimal scale under this scoring function, not a uniquely identified spacing. Other spacings produce comparable localized structure under alternative parameterizations.

4.2 | Phase localization

The 3° model performs better under the tested diagnostics than adjacent alternatives (§4.1); we next ask whether the pattern is uniformly distributed over all possible phases or concentrated at a specific phase. A phase-envelope scan over all origins $\phi \in [0, 3^\circ)$ (300 phases at 0.01° resolution) shows a strong preferred phase: full-corpus A+ counts span 83 to 132 (median 100.0), and Gerizim's phase ($2.2690^\circ \bmod 3^\circ$) sits at the 100.0th percentile (132 A+), the 96.7th for domes (18/90; range 4 to 21). Rayleigh permutation on the 3° -circle: $p_{\text{perm}} = 0.0001$ *** (§5.2). This phase was identified as part of the selected frame (post-selection; §3.3).

4.3 | Cluster Asymmetry (Tests 2 and 3, Primary)

Tests 2 and 3 use the full corpus ($N = 1011$), no keyword filter (§3.5), and quantify alignment without any tier cutoff (Table 3).

Test 2 (Spearman rank correlation between harmonic node mean deviation and site count): $\rho = -0.1076$, permutation $p = 0.00065$ (***). Nodes drawing more sites show smaller mean harmonic deviations. **Test 3** (phase-rotation null): the observed full-corpus mean harmonic deviation is smaller than expected when the reference grid is rotated by a uniform random offset in $[0^\circ, 3^\circ)$ ($p_{\text{perm}} = 0.04729$, *). Both point in the same direction (Holm-adjusted diagnostics: $p_{\text{adj}} = 0.002$ **; $p_{\text{adj}} = 0.047$ *).

The tier-based characterisation below provides descriptive context. For each site, count other UNESCO sites within $\pm 0.30^\circ$ longitude; compare cluster sizes for A+ vs. non-A+ sites (Mann-Whitney U , permutation test). At the harmonic level, compare per-harmonic site counts for A+-bearing vs. non-A+ harmonics.

TABLE 1 | Cluster asymmetry: A+ vs. non-A+ sites in the full UNESCO corpus ($N = 1011$; cluster radius $\pm 0.30^\circ$ lon)

Metric	A+ sites	Non-A+ sites	Ratio / Z	p
Mean cluster size (sites within $\pm 0.30^\circ$ lon)	6.56	4.96	1.32×	
Mann-Whitney U (one-tailed)				0.0025 **
Permutation (10,000 shuffles)			$Z = 3.96$	0.0000 ***
<i>Harmonic-level density (all Tier $\leq B$ sites per 3° harmonic):</i>				
Harmonics with ≥ 1 A+ site (36)	mean = 25.0 sites			
Harmonics with 0 A+ sites (19)	mean = 5.9 sites		4.24×	< 0.0001 ***

A+-bearing harmonics host 4.24× more sites on average than non-A+ harmonics ($p < 0.0001$; Table 1). Sites at harmonic nodes cluster more densely than sites at harmonic midpoints (9.25 vs. 6.49 neighbors; $1.43\times$, $p = 0.0527$).

Subtest: does clustering produce the unit-sensitivity peak?

A joint permutation test (50000 iterations) finds the conditional fraction of clustering-matched permutations producing the canonical-unit peak (0.1667) is identical to the unconditional rate (0.1786): in this check, the clustering and unit-sensitivity summaries appear separable.

Methodological note: Rayleigh phase-lock (Test B). The near-perfect Rayleigh R of the A+ sub-sample is a mathematical consequence of the A+ classification criterion (§6.1) and is reported as a structural description. Three formal permutation tests (10,000 shuffles each, seed 42) address distinct aspects of the periodicity: **Test A (full-corpus Rayleigh, negative control).** Full-corpus Rayleigh $R = 0.0224$ ($Z = 1.06$, $p_{\text{perm}} = 0.3716$, ns): no spurious global 3° periodicity; enrichment is concentrated in A+ sites.

Test B (A+-only Rayleigh, structural check). Among 132 Tier-A+ sites, $R = 0.9842$ ($Z = 23.63$): a structural consequence of the A+ criterion, not independent evidence (§6.1).

Tests C to E (anchor specificity). Circular-shift null: 132 A+ sites vs. null mean 101.05 ($p_{\text{perm}} = 0.0061$, **). Targeted von Mises (full corpus, μ_0 fixed at Gerizim phase): $\hat{w} = 0.0312$ ($\approx 3.1\%$ of sites), $p_{\text{boot}} = 0.0199$ (*). Global anchor sweep (36,000 trial anchors): Levantine corridor $p = 0.0753$ (†).

4.4 | Full-Corpus Enrichment (Test 4, Supporting)

We now quantify the excess at the Gerizim-grid phase.

The full UNESCO Cultural/Mixed corpus ($N = 1011$) shows a Tier-A+ rate of 13.1% (Figure 1) (132/1011; Clopper-Pearson 95% CI [11.0%, 15.3%]), above the 10% null (binomial $p = 0.0011$; supporting Test 4, not in the primary diagnostic family). The observed enrichment is 1.31×, modest relative to the dome subpopulation (2.00×) and the harmonic cluster asymmetry. The primary threshold-free diagnostics are reported in Table 3; they summarize structure within the selected harmonic representation, not independent confirmation.

Region-stratified audit. A region-specific empirical null (Fisher combination of per-region binomials) yields $p = 0.7978$ (ns), so it does not provide region-stratified evidence against geographic concentration. The pre-2000 cohort remains elevated against the geometric null ($p = 0.0132$ *, §4.7), and is treated as descriptive context.

World religion sites

Religion sub-populations are defined by keyword sets documented in the supplementary audit archive (Tenenbaum 2026). A chi-squared test finds no significant A+ rate difference across the five main religion groups ($\chi^2 = 4.753$, $df = 4$, $p = 0.3136$, ns). Both Buddhism (20/71 A-tier, 28.2%; 19/71 C-band, 26.8%; $p = 0.0052$ **) and Hinduism (11/39 A-tier, 28.2%; 10/39 C-band, 25.6%; $p = 0.0324$ *) show a joint A-tier/C-band signal (multivariate hypergeometric). Both persist in tradition-exclusive sites ($p = 0.0123$ *, $p = 0.0862$ †).

4.5 | Domed and Spherical Monuments (Test 5, Supporting)

The keyword filter is described in §3.5 ($N = 90$). The dome sub-corpus should be treated as a hypothesis-generating subpopulation: the keyword set was refined iteratively during analysis, and the enrichment should not be read as confirmatory evidence independent of that process. **Test 1** (dome von Mises mixture, Table 3): a two-component von Mises/uniform mixture with the mean fixed at the Gerizim phase yields $\hat{w} = 0.0876$ ($\hat{\kappa} = 252.19$), $p_{\text{boot}} = 0.00180$ (**; Holm $p_{\text{adj}} = 0.004$, **). Approximately 8.8% of dome sites are concentrated at the Gerizim phase, with an angular spread of $\approx 0.0301^\circ$. The tier-based enrichment below provides complementary descriptive context.

Tier-A++ concentration (supporting result). 11 of 90 dome sites (12.2%, 3.06× the 4% null) fall within Tier-A++ ($\delta \leq 0.06^\circ$, $\lesssim 6.66$ km). This concentration is directionally reproducible across three test statistics: Fisher OR = 2.71, $p = 0.0077$ **; permutation $Z = 3.21$, $p = 0.004$ **; bootstrap $p = 0.006$ **. Dome A++ gives consistent results across all three null models and remains directionally robust under full Bonferroni at $k = 44$ (§5.6).

Tier-A+ enrichment (supporting Test 5). 18/90 sites at Tier-A+ (20.0% [12.3, 29.8], 2.00×; Fisher OR = 1.77, $p = 0.0346$ *, binomial $p = 0.0032$ **). Tier-A: 29/90 (32.2%, 1.61×; Fisher $p = 0.0175$ *, binomial $p = 0.0043$ **). The χ^2 -uniform test ($p = 0.0732$, †) is consistent with concentration at harmonic nodes. Tier-B (middle zone): 45/90 sites; mean $\delta = 0.6586^\circ$.

Tier-A+ sites: Khoja Ahmed Yasawi (0.2 km), Boyana Church (0.3 km), Lumbini (0.8 km), Humayun's Tomb (2.0 km), Pyu Ancient Cities (2.3 km), Silk Roads: Chang'an-Tianshan (2.5 km), Trulli of Alberobello (3.6 km), Old City of Jerusalem (4.1 km), Borobudur (4.3 km), Kizhi Pogost (5.0 km), Echmiatsin/Zvartnots (6.2 km).

A leave-one-out binomial check shows that no single site drives the result: removing any A+ site yields $N = 89$, A+= 17, $p = 0.0067$. Leave-2-out (worst case) across all $\binom{18}{2}$ combinations: $p =$

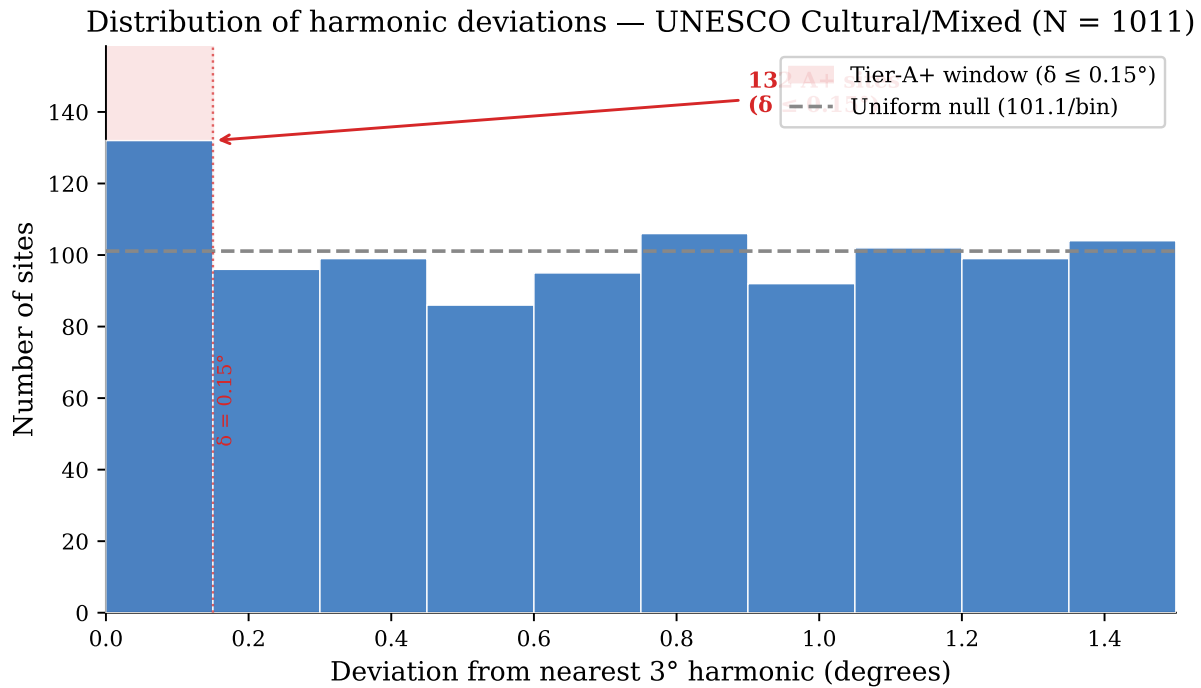


FIGURE 1 | Distribution of harmonic deviations for the full UNESCO corpus. The excess of sites within the Tier-A+ window ($\delta \leq 0.15^\circ$, shaded) above the uniform null (dashed line) illustrates the corpus-level enrichment pattern; Tests 1 and 2 (domed monuments and cluster asymmetry) characterize its morphological and spatial structure.

0.0133. Leave-3-out (worst case): $p = 0.0253$; the evolution corpus (Test 6) (§4.6) and the A++ concentration are more robust. The A++ leave-3-out worst case across all $\binom{11}{3} = 165$ triples yields $p = 0.0233$, remaining below $\alpha = 0.05$ in this audit (leave-2-out worst case $p = 0.0087$).

Multi-scale combined enrichment. A Fisher combination across 15 log-spaced proximity windows (null rates 0.33% to 40%; canonical tiers A++, A+, A are three of the 15) shows all windows enriched above their null (OR $1.2\times$ to $6.7\times$, 14 of 15 individually $p < 0.10$); permutation-calibrated $p = 0.00362$ **. No fixed tier cutoff enters this statistic. Applied jointly across all four corpora (UNESCO domes, UNESCO full, Wikidata Q180987 stupas, OWTRAD Silk Road vertices), the maximum joint enrichment falls at approximately 29 km from a harmonic node (1/10th of the ~289 km inter-node spacing at 30°N), with permutation-calibrated $p = 0.01232$ *. Each corpus peaks within this 29 km bound (4 to 29 km across corpora), a scale that emerges from the data rather than any pre-specified threshold.

Context-validated robustness (Test 5x). Sentence-level validation ($N = 83$, 7 rejected): A++ $p = 0.0041$ **; A+ $p = 0.0614$ †. The A++ result strengthens when false positives are removed; A+ remains directionally elevated under this stricter filter. Keyword-stratified check: stupa/stupas alone ($n = 18$) carries the strongest signal (A++ $p = 0.0057$ **); dome/tholos/spherical without stupa ($n = 65$) remains elevated ($p = 0.0193$ *).

Geographic-concentration checks (dedicated simulation). Domed monuments are concentrated in the Eurasian corridor: 71% fall in 20° to 120°E versus 41% of the full corpus. Two targeted resampling checks evaluate whether this explains the dome enrichment (100,000 trials).

Null A (within-dome bootstrap stability check): Draw $N = 90$ longitudes from the dome sites' own list. Mean = 18.00 A+,

$p = 0.5429$ (ns). The dome sites' positions are concentrated near harmonic longitudes; this check characterizes the observed dome distribution rather than serving as an external geographic null.

Null C (restricted geographic-concentration null): Pool all full-corpus sites within $\pm W^\circ$ of any dome site and bootstrap the expected A+ count. Dome sites (18 A+, 20.0%) exceed the broader footprint at all three windows tested: $\pm 2^\circ$ ($p = 0.0257$, *); $\pm 5^\circ$ ($p = 0.0393$, *); $\pm 10^\circ$ ($p = 0.0442$, *). Results are stable across the tested window widths.

In the global anchor sweep (§5.2), Gerizim scores 132 A+ (99.7th percentile); Jerusalem scores 128 (97.7th).

4.6 | Hemispherical Mound Evolution (Test 6, Sensitivity)

Extending the dome keyword subset of Test 5 with earthen mound-stage keywords (see §3.5); the combined population ($N = 110$) adds 20 mound-stage sites to the core domed/spherical population, with 4 sites matching both keyword sets.

Stage-by-stage breakdown (Table 2):

The combined population ($N = 110$, A+ = 18.2%; Table 2) is consistent with Test 1. The mound stage does not reach significance (8.3%, OR = 0.60 $\times$, Fisher $p = 0.8430$, ns). The signal is carried by the stupa (A+ 33.3%, OR = 3.44 $\times$, $p = 0.0216$ *) and dome sub-stages (A+ 17.3%, OR = 1.44 $\times$, $p = 0.1663$ ns). Leave-one-out checks confirm stability. Leave-3-out (worst case, 1140 triples): $p = 0.0371$.

Stupa geographic-concentration null. Stupa sites are concentrated in South/Southeast Asia (72% in 70° to 110°E). Null A (within-stupa bootstrap): $p = 0.5879$ (ns). Null C (restricted $\pm 5^\circ$ draw): $p = 0.0556$ (†).

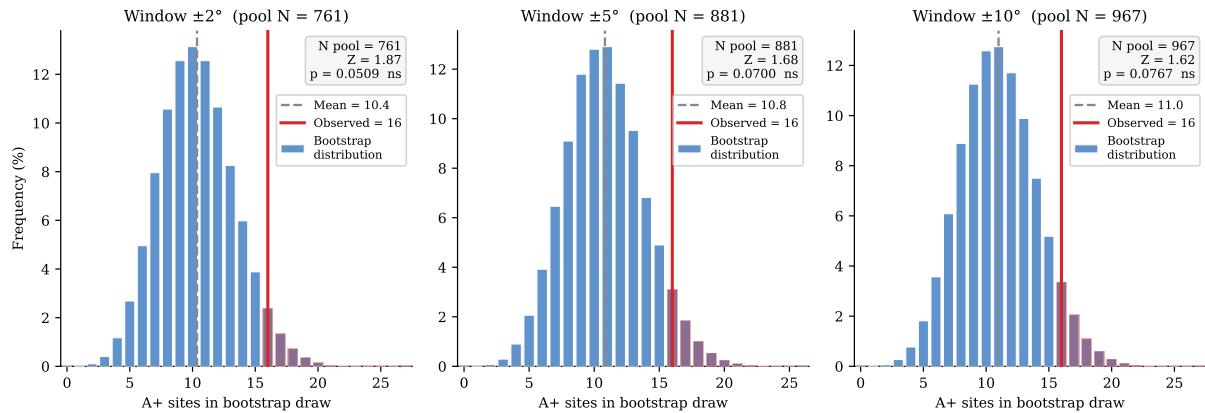


FIGURE 2 | Null C bootstrap distributions for the dome sub-population ($N = 90$, observed $A+ = 18$) at three window widths. For each window $\pm W^\circ$, all full-corpus sites within $\pm W^\circ$ of any dome longitude form the restricted pool; 10^5 draws of size $N = 90$ from that pool define the expected $A+$ distribution (blue bars). The observed count (red line) exceeds the pool mean by $Z \approx 2.7$ in all three windows ($\pm 2^\circ$: $p = 0.0257$, *; $\pm 5^\circ$: $p = 0.0393$, *; $\pm 10^\circ$: $p = 0.0442$, *), showing stability across the tested window widths.

TABLE 2 | Test 6 enrichment by evolutionary stage ($N = 110$). Null rates: $A++ P_0 = 4\%$, $A+ P_0 = 10\%$, $A P_0 = 20\%$. Fisher exact tests: stage vs. full corpus background.

Stage	N	Tier A++ (≤ 6.66 km)		Tier A+ (≤ 16.6 km)		Tier A (≤ 33 km)	
		A++	Rate OR (p)	A+	Rate OR (p)	A	Rate OR (p)
Mound ^a	24	2	8.3% 1.57 $\times$ (0.3880 ns)	2	8.3% 0.60 $\times$ (0.8430 ns)	6	25.0% 1.15 $\times$ (0.4653 ns)
Stupa ^b	18	4	22.2% 5.17 $\times$ (0.0144 *)	6	33.3% 3.44 $\times$ (0.0216 *)	7	38.9% 2.22 $\times$ (0.0874 †)
Dome ^c	75	8	10.7% 2.21 $\times$ (0.0485 *)	13	17.3% 1.44 $\times$ (0.1663 ns)	23	30.7% 1.58 $\times$ (0.0576 †)
Combined	110	13	11.8% 2.67$\times$ (0.0658 **)	20	18.2% 1.57$\times$ (0.0658 †)	35	31.8% 1.71$\times$ (0.0113 *)

^atumulus/barrow/kofun. ^bstupa/stupas. ^cdome/tholos/spherical. OR = odds ratio; p = one-sided Fisher exact vs. full UNESCO cultural corpus ($N = 1011$). Stage N values sum to more than the combined total because 4 sites match multiple stages.

4.7 | Temporal Gradient (Test 7, Descriptive)

The Tier-A++ rate in the founding canon (1978 to 1984) is 8.0% (11/138; 1.38 $\times$ the 4% null, $p = 0.0232$ *), declining monotonically toward the null in later cohorts (2000 to 2025: 4.5%). Cochran-Armitage three-cohort trend test (A++, decreasing): $Z = -1.664$, $p = 0.0481$ (*). The five-cohort trend is marginal ($p = 0.0489$ *). By contrast, the A+ rate shows no significant decreasing trend ($p = 0.4184$ ns), consistent with the gradient being specific to the highest-precision tier. This is consistent with UNESCO's founding canon over-representing historically prominent monuments, some of which are also precisely aligned to harmonic nodes. The Mantel-Haenszel stratified test is under-powered at the sub-group level; Logistic regression with region as a covariate is directionally consistent with the A++ gradient ($Z = -1.760$, $p = 0.0784$ †, OR = 0.722). The declining rate may partly reflect UNESCO's expansion into regions where hemispherical traditions are less prevalent. The gradient is treated as descriptive context only (Figure 3).

4.8 | FDR audit and Bonferroni correction

Results are consolidated in Table 3. A comprehensive FDR audit (44 tests; Benjamini & Hochberg 1995) retains 22 at $q < 0.05$; 6 remain below the full-Bonferroni audit threshold at $k = 44$: dome/spherical A++ ($p = 0.0416$), cluster permutation ($p = < 0.0001$), harmonic Mann-Whitney ($p = 0.0012$), sequential-peak

permutation ($p = 0.0279$), and evolution combined A++ ($p = 0.0208$).

5 | Robustness Checks

5.1 | Unit Sweep and Sensitivity Slope

Unit sweep and sensitivity slope results are in §4.1. Briefly: nearby non-harmonic spacings are non-significant; a $\pm 0.3\%$ shift collapses joint significance; Monte Carlo sweep (§6.2) supports the relative peak; Tests C and D (§4.3) support phase specificity within the selected 3° frame.

5.2 | Global Anchor Sweep

The global sweep (36,000 trial anchors, 0.01° resolution) uses the full corpus plus synthetic Gerizim (§3.2), with each anchor self-excluding ($N = 1011$ per trial).

Both anchors place the corridor in the global top 2.3% (36,000 trial anchors, 0 to 360° E; Gerizim: 132 A+, 99.7th percentile; Jerusalem: 128 A+, 97.7th percentile). The result is anchor-invariant within the 4.1 km corridor window. Other historical meridians tested here do not show comparable enrichment.

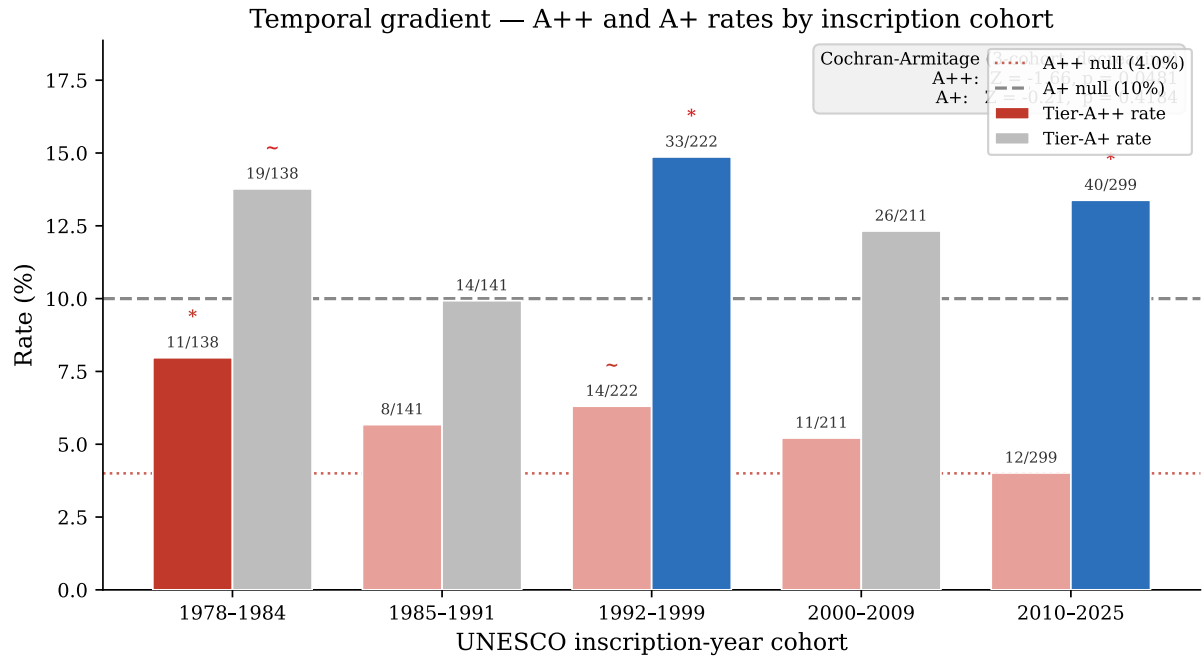


FIGURE 3 | Tier-A++ and Tier-A+ rates by UNESCO inscription-year cohort. The founding canon (1978 to 1984) shows an elevated A++ rate (8.0%, $p = 0.0232$ *) that declines monotonically across later cohorts toward the 4% null (dotted line; Cochran-Armitage $Z = -1.664$, $p = 0.0481$ *). The A+ rate (right bars) shows no significant trend ($p = 0.4184$ ns).

TABLE 3 | Multiple-comparisons summary. Primary diagnostic family: Tests 1 to 3 (threshold-free, $k = 3$). Supporting enrichment tests (Tests 4 and 5) use a pre-specified $p_0 = 10\%$ proximity window; see §3.5.

Test	N	Raw p	Holm p_{adj} , $k=3$	Status
Test 1 (dome VM mix.)	90	0.00180 ^b ($w = 0.0876$, $\kappa = 252.19$)	0.004 **	Diagnostic
Test 2 (Spearman)	1011	0.00065 ^b ($\rho = -0.1076$)	0.002 **	Diagnostic
Test 3 (mean dev.)	1011	0.04729 ^b	0.047 *	Diagnostic
Test 4 (full-corpus A+)	1011	0.0011 ^a	N/A	Supporting
Test 5 (dome A+)	90	0.0032 ^a (OR = 1.77)	N/A	Supporting
Test 6 (mound evol.)	110	0.0658 ^a (OR = 1.57)	N/A	Sensitivity
Test 7 (temporal)	1011	0.4184 ^b	N/A	Descriptive

^aBinomial or Fisher exact p vs. 10% geometric null (window = 0.15°; convenience choice, sensitivity-audited over 2% to 10% null rates in Supplementary S1);
^bpermutation or bootstrap p . Tests 1 to 3 are threshold-free; no tier cutoff enters any primary diagnostic. Adjusted p -values describe calibration within the selected representation, not confirmatory rejection of a global discovery null. Holm (1979) step-down (Holm 1979).

The $x.25^\circ\text{E}$ phase band

Every $x.25^\circ\text{E}$ longitude (120 anchors) produces identically the maximum A+ count, spaced 3° apart. The flat-top width is a function of the tier threshold chosen: A++ ($\pm 0.06^\circ$), A+ ($\pm 0.15^\circ$), A ($\pm 0.3^\circ$); the number of tied anchors is a methodological artifact, not a physical feature. Mount Gerizim is a fixed landmark at 35.269°E (phase 2.269° , gap $0.019^\circ \approx 1.8$ km). It lies within the A and A+ flat-top bands but not A++.

Formal statistical tests for the $x.25^\circ\text{E}$ phase band

The full-corpus Rayleigh ($p_{\text{perm}} = 0.3716$, ns) shows no global 3° spectral periodicity; Test B is tautological. Tests C ($p_{\text{perm}} = 0.0061$, **) and D ($p_{\text{boot}} = 0.0199$, *) are positive anchor-specific tests; Test E ($p = 0.0753$, †, marginal) is the anchor discovery correction. The unit sensitivity sweep (§5.1) and Monte Carlo unit sweep (§6.2) identify 3° as the highest-scoring spacing under the

template score, given the pre-specified Gerizim anchor; neither sweep constitutes anchor-free or period-free discovery.

Corridor and landmark ranking. Of the 13 A+ sites unique to Gerizim and 9 unique to Jerusalem, 119 are shared; the enrichment belongs to the corridor, not to either anchor individually. Using each UNESCO site as anchor, Gerizim ranks #8 among the extended candidate landmarks (97th percentile).

5.3 | Spatial Autocorrelation

A+ harmonics host $4.24\times$ as many total UNESCO sites as non-A+ harmonics ($p < 0.0001$). DEFF correction via ICC ($\pm 0.5^\circ$, $\rho = 0.187$): DEFF = 1.842, $N_{\text{eff}} = 549$, corrected $p = 0.011$. Block bootstrap (3° blocks, 100,000 resamples): 95% CI

[10.50%, 15.43%] excludes the 10% null. Within-group permutation is non-discriminating for alignment vs. geographic concentration; the restricted geographic-concentration null (§4.5) addresses that comparison directly.

5.4 | Null Model Validation

Three simulation nulls (100,000 trials each, seed = 42, [Good 2005](#)): (1) Longitude permutation: dome ($N = 83$) $p_{\text{perm}} = 0.060^{\dagger}$; 1978-1984 canon ($N = 138$) $p = 0.433$ ns; pre-2000 ($N = 501$) $p = 0.490$ ns. (2) Bootstrap from full corpus: dome $p_{\text{boot}} = 0.070^{\dagger}$. (3) KDE-smoothed null: expected 101.1 ± 9.6 , observed 132, $Z = 3.23$, $p = 0.001^{**}$. The KDE null ($p = 0.001^{**}$) supports dome enrichment; permutation and bootstrap are marginal at the A+ level but stronger for the A++ sub-population (permutation $p = 0.004^{**}$, bootstrap $p = 0.006^{**}$). The within-dome bootstrap characterizes the observed concentration near harmonic longitudes, while the restricted geographic-concentration null (Null C; §4.5) indicates that longitude footprint alone is not sufficient to explain the dome enrichment.

5.5 | Pipeline Portability: Wikidata Q180987 Stupa Corpus

The pipeline was applied to $N = 229$ Wikidata-attested stupas (class Q180987) in a separately assembled corpus ([Tenenbaum 2026](#)). The Kedu Plain (Java) hosts the densest cluster at the 75° harmonic (75°E from Gerizim), anchored by Borobudur (c. 800 CE, 7.23 km, Tier A+) ([Miksic 1990](#)); the World Peace Pagoda at Lumbini achieves A++ precision (~555 m) under the Gerizim reference frame.

Using the identical anchor and null, the corpus yields 70/229 Tier-A sites (30.6%, $1.53\times$ null; binomial $p < 0.001^{***}$; permutation $Z = 1.59$, $p_{\text{perm}} = 0.065$, ~). No Tier-A+ enrichment is found (binomial $p = 0.021$, *), consistent with cluster-level rather than individual-site precision. Harmonic nodes with at least one Tier-A site carry $2.56\times$ more stupa sites on average than empty nodes, mirroring the cluster-asymmetry result of Test 3.

Geographic tier breakdown and bimodal structure. All regional p-values below are one-sided Fisher exact tests. A sub-regional breakdown reveals a bimodal pattern consistent with the Buddhist and Hindu bimodal enrichment in the UNESCO religion audit (§4.4). The Java/Sumatra node (105° to 115°E, near the 75° harmonic at 110.3°E) concentrates 43 of 83 sites at Tier-A (51.8%, OR = $4.74\times$, $p < 0.001^{***}$), rising to 42/54 (77.8%, OR = $18.38\times$, $p < 0.001^{***}$) in the 107° to 112°E window, where A+ is also elevated at 31.5% (OR = $4.57\times$, $p < 0.001^{***}$). The physical heartland, India/Nepal (70° to 90°E) and Myanmar/Cambodia (90° to 105°E), is A-tier depleted (combined OR = $0.21\times$, $p < 0.001^{***}$, one-sided depletion test) but strongly enriched in the inter-harmonic C-band: Myanmar/Cambodia reaches 35.8% (OR = $4.47\times$, $p < 0.001^{***}$) and the combined heartland 29.2% (OR = $6.00\times$, $p < 0.001^{***}$). This A/C anti-correlation mirrors the bimodal pattern in the religion audit (§4.4). Non-heartland Wikidata stupas (outside 70° to 110°E, $N = 102$) show the same A-enriched, C-depleted profile: 50.0% reach Tier-A (OR = $5.68\times$, $p < 0.001^{***}$) with no C-band enrichment (OR = $0.19\times$, $p = 1.000$ ns).

5.6 | Multiple Comparisons

See §3.6 for the full protocol. Gerizim ranks at the 99.7th percentile; a $\pm 0.3\%$ unit shift collapses joint significance. FDR audit (44 tests) retains 22 at $q < 0.05$; 6 remain below the Bonferroni audit threshold at $k = 44$.

6 | Limitations

6.1 | Exploratory Status and Multiple-Comparisons Exposure

The analysis is exploratory: spacing, phase, anchor, sub-corpus, and tier thresholds were all examined iteratively on a public dataset for which prospective constraint is not applicable. The Holm correction covers only the three primary diagnostics; p-values are descriptive, not confirmatory. They are not tests of discovery. They quantify how sharply the selected representation describes the data. Key degrees of freedom (3° spacing, Gerizim phase, keyword set, tier thresholds) have external motivation; threshold sensitivity is in Supplementary S1. External corpora are not independent of UNESCO (§7.3).

6.2 | Geographic and Cultural Selection Bias

The UNESCO list over-represents Europe and the Mediterranean-to-South-Asian corridor (Europe/N. America 50%; Arab States 9%; Asia-Pacific 23%; Latin America 11%; Africa 7%). Domed monuments exceed the A+ rate of other monuments from the same longitude footprint (Null C, $p = 0.0393$, *), which indicates that architectural type may contribute to the enrichment beyond longitude footprint alone.

A Monte Carlo unit sweep (10000 permutations, 18 candidate spacings 0.5° to 15°) gives the highest template score at 3° ($p < 0.001$), with harmonics at 6° and 12° also elevated under the same score. The dome subset yields 1 independent high-scoring spacings ($P = 0.2248$ under the geographic null).

6.3 | Representation Dependence

The observed structure depends on the representation. It is recovered under nearest-harmonic-deviation scoring but not by invariant spectral methods (no Rayleigh peak at 3° in any sub-corpus). Three converging diagnostics: (i) a $\pm 0.3\%$ unit perturbation collapses joint significance (§4.1); (ii) multi-scale enrichment is broad, not a sharp spectral line; (iii) the anchor-sweep maximum is marginal ($p = 0.0753$, †) while targeted tests at the identified frame are stronger (§5.2). The pattern appears under the representation, not as intrinsic periodicity. It arises from the interaction between the scoring function and the imposed coordinate frame. A clustered control shows that the unconstrained global maximum over all (ω, ϕ) is typical under realistic dome-site clustering ($p = 0.522$, $N_{\text{sim}} = 2000$) and does not uniquely recover the hypothesised frame.

6.4 | Americas Negative Control

The UNESCO Americas sub-corpus ($N = 145$) shows A+ rate 9.7%, below the 10% null ($p = 0.5149$ ns, one-sided depletion).

6.5 | Other Limitations

Geocoding precision. UNESCO XML coordinates are administrative centroids (unsystematic noise, no directional bias).

Metrological evidence. No ancient surveying instrument calibrated to metrological sub-units has been recovered; the bēru (Hunger & Steele 2019, Powell 1987) is noted as historical coincidence. All thresholds are degree-denominated.

7 | Discussion

7.1 | Template Alignment vs. Intrinsic Periodicity

The present results are best treated as *template alignment* (§6.3): a reproducible, phase-localized pattern appears under the representation but is not invariant and does not imply a periodic generative model. The analysis does not demonstrate an underlying spatial law or intentional placement. It shows that coordinate-aligned statistics can produce reproducible deviations in clustered datasets. Interpretation should focus on properties of the representation, not the existence of a latent periodic structure.

7.2 | Alternative Explanations

(1) Independent convergence. Hemispherical structures recur across cultures (Aveni 2001 2003), but convergence predicts no longitude specificity.

(2) Geographic concentration. Null C: other monuments from the same longitude footprint are less enriched than the dome subset ($p = 0.0393$, *); these checks reduce, but do not eliminate, the possibility that enrichment partly reflects corridor concentration.

(3) Random coincidence. Simulation nulls (KDE, permutation, bootstrap) support dome enrichment within the selected representation, most strongly at A++ (permutation $p = 0.004$ **, full results in §5.4). Tests C and D support phase specificity within the selected frame (§4.3).

7.3 | Cross-Corpus Consistency: OWTRAD and Wikidata

The same representation-specific pattern appears in the OWTRAD route network (Ciolek 2004) and Wikidata Q180987 stupa inventory, but these share substantial geographic and cultural structure with UNESCO (overlapping corridor, monument classes, and site lists) and are not statistically independent. Their agreement is cross-corpus consistency under a shared representation, not independent replication.

OWTRAD route network. The OWTRAD route network contains 1946 route segments and 1674 unique endpoints. Under the same representation, the degree-weighted endpoint distribution shows A++ enrichment (193 of 3498 endpoint positions, $p = < 0.0001$ ***), while the deduplicated endpoint distribution is weaker (84/1674, $p = 0.0224$ *). A phase-shift test gives $p =$

0.0202 *, and a cluster-asymmetry check gives $p = 0.0007$ ***. These results are treated as shared-representation consistency, not independent replication. The Silk Road corridor is illustrated in Figure 5; harmonic tier distributions for both OWTRAD populations are shown in Figure 4.

8 | Conclusion

This paper reports an exploratory pattern: the UNESCO Cultural and Mixed corpus shows a consistent template-alignment excess near the $\sim 35^\circ\text{E}$ corridor under multiple diagnostic checks (Table 3; $p_{\text{adj}} = 0.004, 0.002, 0.047$), with small effect sizes and corridor-level rather than point localization. These p -values describe within-frame consistency, not confirmatory evidence.

A multi-scale combined test reduces dependence on any single tier choice (dome subset $p = 0.00362$ **; joint enrichment peaks near 29 km, $p = 0.01232$ *). The Wikidata stupa corpus and OWTRAD route network show similar enrichment patterns under the same reference frame (§7.3).

In short, a harmonic-alignment statistic identifies a weak but reproducible pattern under a specific coordinate representation. This pattern is not invariant, does not support a periodic generative model, and is consistent with the interaction between geographic clustering and the imposed harmonic frame. Its primary contribution is methodological. It shows how representation choice can induce stable but non-invariant spatial patterns in culturally clustered datasets. Independent re-examination on corpora without UNESCO's Eurasian sampling structure is the appropriate next step.

APPENDIX

A | Von Mises Mixture Characterisation of External Corpora

A von Mises + uniform mixture model $f(\theta|\kappa) = w \cdot \text{VM}(0, \kappa) + (1 - w) \cdot \text{Uniform}(0, 2\pi)$ was fitted to the harmonic phase deviations ($\theta \in [0, \pi]$, $\theta = 0$ on-harmonic) of two external corpora using maximum likelihood with a grid of starting values.

OWTRAD Silk Road nodes ($N = 1674$): $\hat{\kappa} = 9.58$, $\hat{w} = 0.0229$, $\hat{\sigma} = \kappa^{-1/2} = 0.1543^\circ$.

Wikidata Q180987 stupas ($N = 229$): $\hat{\kappa} = 6.17$, $\hat{w} = 0.1297$, $\hat{\sigma} = 0.1922^\circ$.

N -weighted pooled $\hat{\sigma} = 0.1588^\circ$; κ ratio = 1.6 across corpora.

These values characterize external-corpus clustering and are not used to derive tier boundaries; agreement with tier widths is descriptive context.

B | Tier-Threshold Sensitivity Audit

A log-scale null-rate sweep across 73 thresholds supports the interpretation that enrichment is not an artifact of the specific tier width: 57 of 73 thresholds reach $p < 0.05$ for the dome subset; dome OR ranges $1.53\times$ – $3.40\times$ across the three canonical tiers (A, A+, A++). Within ± 1 log-decade of the A+ threshold, 27 of 28 thresholds are significant. Full results in `supplementary/audit/tier_sensitivity_audit.txt`.

OWTRAD Silk Road network — harmonic deviation distribution (degrees)
 Left: each city once. Right: each city weighted by its route-count.
 Shaded: A++ / A+ / A windows. Dashed: uniform null expectation.

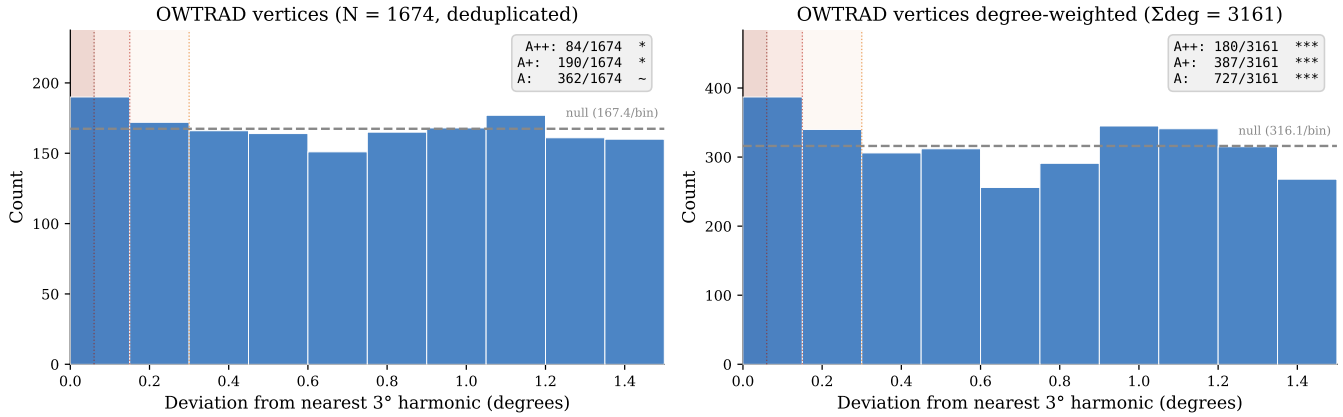


FIGURE 4 | OWTRAD Silk Road endpoint tier distributions: 1674 deduplicated endpoints (left) and 3498 degree-weighted positions (right). The degree-weighted distribution shows A++ enrichment ($z = 4.58$, $p = < 0.0001$ ***); the deduplicated endpoint distribution does not ($z = 2.13$, $p = 0.0224$ *).

Silk Roads corridor — UNESCO dome/stupa and Wikidata Q180987 stupa tier distribution
 Circles = UNESCO (N = 90); triangles = Wikidata (N = 229). Dashed grid: 3° harmonics. Crimson = A+, orange = A-tier.

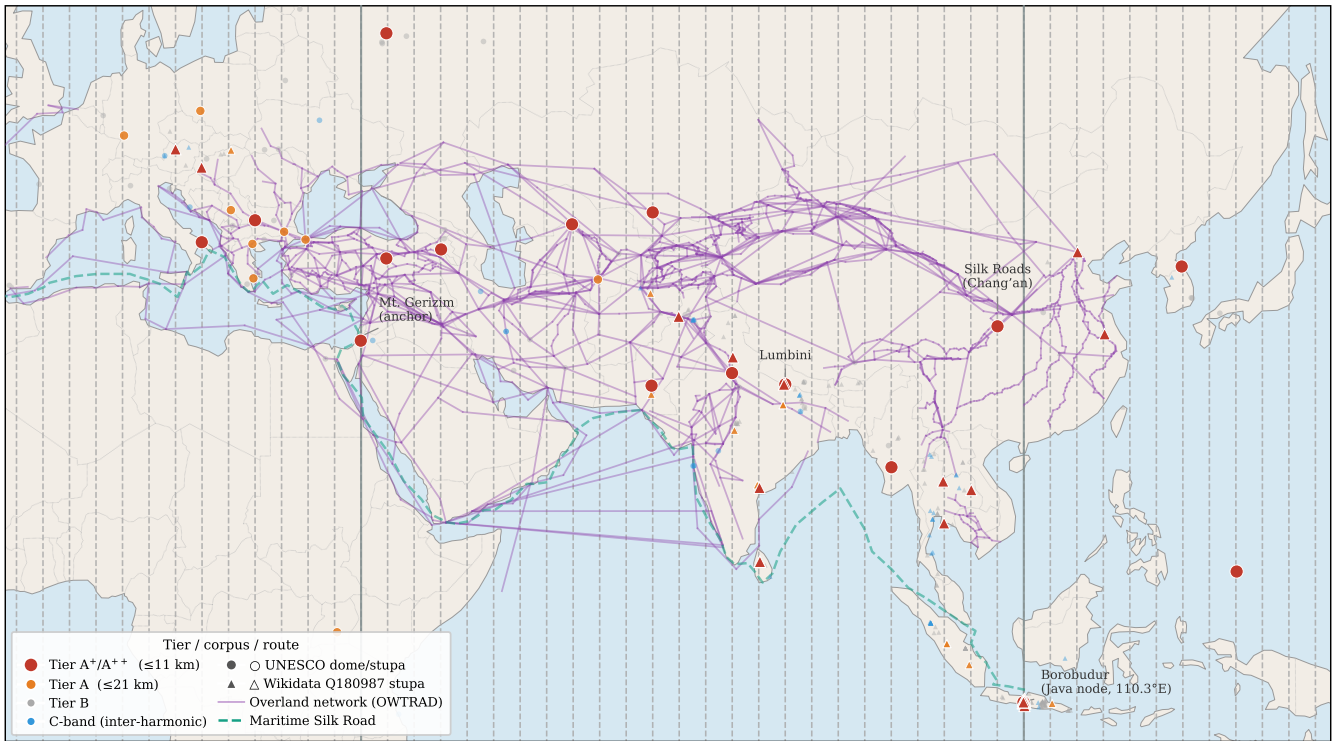


FIGURE 5 | Tier distribution of UNESCO domed/stupa and Wikidata Q180987 stupa sites along the Eurasian corridor. Overland route edges from the OWTRAD attested-route database (Ciolek 2004). Maritime route is a schematic coastal polyline after Casson (1989), Sen (2014), and Tomber (2008). An interactive version is provided as supplementary material (supplementary/interactive_map/silkroad_interactive.html).

C | Keyword Audit Summary

Full site-by-site audit files are indexed in Supplementary S1 (supplementary/Supplementary_S1.md; Tenenbaum 2026); regenerate all with `bashtools/update_all_audits.sh`.

C.1 | Dome/Spherical Monument Filter (Tests 1, 5, 5x, 6x)

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV

text. All 90 matching sites are included in the Test 1 dome mixture and Test 5 tier-enrichment analyses without context gating. For Test 5x, ambiguous keywords (*dome, domed, domes, spherical*) are subject to a two-gate context check: Gate 1 (28 negative patterns) rejects non-architectural uses; Gate 2 (86 positive patterns) requires architectural confirmation. Unambiguous keywords (*stupa/stupas, tholos*) are accepted without validation. The seven rejected sites are: Hiroshima Peace Memorial, Richtersveld, Gyeongju, Viñales Valley, Rock Islands Southern Lagoon, Uluru-Kata Tjuta, and Diquis Stone Spheres.

Test 6 adds earthen mound-stage keywords (*tumulus, tumuli, barrow, barrows, kofun* unambiguous; *mound* context-validated) to trace the hemispherical tradition from burial mound through stupa to architectural dome. Test 6x applies the same architectural context gates to the extended population. Statistics for all four variants are reported in §4.5 and §4.6.

C.2 | Founding-Origin Classifier (Post-Hoc)

Results. The classifier finds 31.8% of A+ sites carry a founding keyword versus 28.7% in the full corpus (Fisher OR = 1.19, $p = 0.225$ ns). The A++ sub-population (56 sites) shows a stronger signal: 42.9% carry a founding keyword (OR = 1.94, $p = 0.014$ *), consistent with the most precisely aligned sites being disproportionately historically foundational monuments. The classifier assigns sites to five founding-related categories; full keyword definitions and the site-by-site listing are in the supplementary archive (Tenenbaum 2026).

References

- Aveni, A. F. (2001). *Skywatchers: A Revised and Updated Version of Skywatchers of Ancient Mexico*. University of Texas Press.
- Aveni, A. F. (2003). Archaeoastronomy in the ancient Americas. *Journal of Archaeological Research*, 11(2), 149–191.
- Casson, L. (1989). *The Periplus Maris Erythraei: Text with Introduction, Translation, and Commentary*. Princeton University Press.
- Ciolek, T. M. (2004). *Old World Trade Routes (OWTRAD) Project*. Asia Pacific Research Online, Canberra. <http://www.ciolek.com/OWTRAD/DATA/tmcOWTRAD.html>
- Benjamini, Y. & Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society, Series B*, 57(1), 289–300.
- Crown, A. D. (1989). *The Samaritans*. J. C. B. Mohr (Paul Siebeck).
- Freeman, P. R. (1976). A Bayesian analysis of the megalithic yard. *Journal of the Royal Statistical Society, Series A*, 139(1), 20–35.
- Good, P. I. (2005). *Permutation, Parametric, and Bootstrap Tests of Hypotheses* (3rd ed.). Springer.
- Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70.
- Hansen, V. (2012). *The Silk Road: A New History*. Oxford University Press.
- Harvey, P. D. A. (1987). Medieval maps of the world. In J. B. Harley & D. Woodward (Eds.), *The History of Cartography, Volume 1: Cartography in Prehistoric, Ancient, and Medieval Europe and the Mediterranean* (pp. 283–370). University of Chicago Press.
- Burton, R. F. (1893). *Personal Narrative of a Pilgrimage to Al-Madinah & Meccah*, vol. 2 (Memorial Edition), p. 63, n. 2. Tylston and Edwards.
- Hunger, H. & Steele, J. (2019). *The Babylonian Astronomical Compendium MUL.APIN*. Routledge.
- Thureau-Dangin, F. (1921). *Numération et métrologie sumériennes*. *Revue d'Assyriologie et d'archéologie orientale*, 18, 123–142.
- Kendall, D. G. (1974). Hunting quanta. *Philosophical Transactions of the Royal Society A*, 276(1257), 231–266.
- MacDonald, J. (1964). *The Theology of the Samaritans*. SCM Press.
- Mardia, K. V. & Jupp, P. E. (2000). *Directional Statistics*. Wiley.
- Magli, G. (2013). *Architecture, Astronomy and Sacred Landscape in Ancient Egypt*. Cambridge University Press.
- Magli, G. (2016). *Archaeoastronomy: Introduction to the Science of Stars and Stones*. Springer.
- Miksic, J. N. (1990). *Borobudur: Golden Tales of the Buddhas*. Periplus Editions.
- Neugebauer, O. (1975). *A History of Ancient Mathematical Astronomy*, 3 vols. Springer.
- Powell, M. A. (1987). Masse und Gewichte. In D. O. Edzard (Ed.), *Reallexikon der Assyriologie und Vorderasiatischen Archäologie*, vol. 7, pp. 457–517. Walter de Gruyter.
- Purvis, J. D. (1968). *The Samaritan Pentateuch and the Origin of the Samaritan Sect*. Harvard University Press.
- Robson, E. (2008). *Mathematics in Ancient Iraq: A Social History*. Princeton University Press.
- Ruggles, C. L. N. (1999). *Astronomy in Prehistoric Britain and Ireland*. Yale University Press.
- Ruggles, C. L. N. & Cotte, M. (Eds.) (2010). *Heritage Sites of Astronomy and Archaeoastronomy in the Context of the UNESCO World Heritage Convention: A Thematic Study*. ICOMOS & International Astronomical Union.
- Ruggles, C. L. N. (Ed.) (2015). *Handbook of Archaeoastronomy and Ethnoastronomy*. Springer.
- Sen, T. (2014). The spread of Buddhism. In B. Z. Kedar & M. E. Wiesner-Hanks (Eds.), *The Cambridge World History, Vol. 5: Expanding Webs of Exchange and Conflict, 500 CE–1500 CE* (pp. 447–479). Cambridge University Press.
- Silva, F. (2020). A probabilistic framework and significance test for the analysis of structural orientations in skyscape archaeology. *Journal of Archaeological Science*, 118, 105138.
- Tenenbaum, S. (2026). *Longitude Quantization in the UNESCO World Heritage Corpus: Domes, Stupas, and 3° Harmonics: Code and Data* (v1.6.0). Zenodo. <https://doi.org/10.5281/zenodo.19938283>
- Tomber, R. (2008). *Indo-Roman Trade: From Pots to Pepper*. Duckworth.
- Thom, A. (1964). The larger units of length of megalithic man. *Journal of the Royal Statistical Society, Series A*, 127(4), 527–533.
- Thom, A. (1967). *Megalithic Sites in Britain*. Oxford University Press.
- Thom, A. (1971). *Megalithic Lunar Observatories*. Oxford University Press.
- UNESCO World Heritage Centre. (2024). *World Heritage List* [XML dataset]. Retrieved from <https://whc.unesco.org/en/list/xml/>.
- Watson, R. T. & Horowitz, W. (2011). *Writing Science Before the Greeks: A Naturalistic Analysis of the Babylonian Astronomical Treatise MUL.APIN*. Brill.
- Williamson, T. & Bellamy, L. (1983). *Ley Lines in Question*. World's Work.

DATA AVAILABILITY STATEMENT

All data and code are openly available; a permanent snapshot (v1.6.0) is deposited at Zenodo (Tenenbaum 2026; <https://doi.org/10.5281/zenodo.19938283>); UNESCO XML: UNESCO World Heritage Centre 2024. Scripts, audit files, and maps are indexed in Supplementary S1 (supplementary/Supplementary_S1.md), with each file's role, macros produced, and paper section noted. Full reproduction: `reproduce_all_macros.sh`, `update_all_audits.sh`, `generate_figures.py`.

ACKNOWLEDGEMENTS

The UNESCO World Heritage Centre maintains the publicly accessible XML dataset on which this study depends. This research was conducted independently and without external institutional affiliation.

CONFLICT OF INTEREST

The author declares no conflict of interest.

SOFTWARE AND AI DISCLOSURE

All statistical analyses were performed in Python 3 using NumPy, SciPy, and pandas. Large-language-model tools were used during preparation of this manuscript: Claude (Anthropic) assisted with prose drafting and editing; GitHub Copilot (Microsoft) assisted with code autocompletion. These tools were used solely as writing and coding aids. All scientific hypotheses, analytical choices, keyword-list decisions, interpretation of results, and conclusions are solely the responsibility of the author. No AI tool was used to generate, fabricate, or substitute for primary data or statistical analysis.